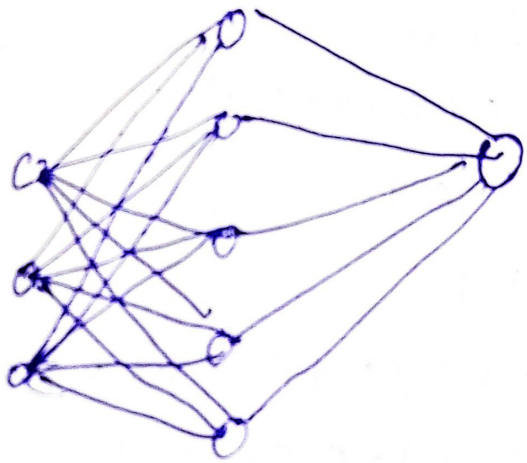
Tutorial - 2 . 2403A52014

An e-commerce platform classifies reviews as positive or Negetative using a single layer neural network, and hidden layer with 5 neurons.

x_1	x_2	x_3	Target
6	1	1	1
2	4	1	0
5	2	1	1
1	3	1	0

Initialize the weights $w_1 = 0$, $w_2 = 0$, $w_3 = 0$, $b = 1$

For sample 1



$$net = x_1 w_1 + x_2 w_2 + x_3 w_3$$

$$y = \begin{cases} 1, & net \geq 0 \\ 0, & net < 0 \end{cases}$$

init $w_1 = w_2 = w_3 = 0$

$net = 0 \times 6 + 0 \times 1 + 0 \times 1$

$net = 0 \quad y = 1 \quad \hat{y} = 1$

$e = y - \hat{y} = 1 - 1 = 0$

No weights update required.

For sample 2 :

$net = 0.2 \times 0.4 \times 0.1$

$net = 0 \cdot \hat{y} = 1 \quad y = 0$

$e = y - \hat{y}$

$e = -1.$

$w_1(new) = w(old) + \eta * e(x_1)$

$= 0 + 0.1 \times (-1)(2)$

$= 0.1 \times (-2)$

$= -0.2$

$w_2(new) = 0 + 0.1 \times (-1)4$

$= -0.4.$

$w_3(new) = 0 + 0.1 \times -1(1)$

$= -0.1.$

For sample 3 :

$net = -0.2 \times 5 + (-0.4) \times 2 + (-0.1) \times 1$

$net = -1 - 0.8 - 0.1$

$= -1.9 \quad \hat{y} = 0, \; y = 1$

$e = y - \hat{y}$

$e = 1.$

$w_1(new) = -0.2 + 0.1 \times (1)(5)$

$= 0.3.$

$w_2(new) = -0.4 + 0.1 \times (1)(2)$

$= -0.2$

$w_3(new) = -0.1 + 0.1 \times (1)(1)$

$= 0$

-.03

For sample 4

$$net = 0.3 \times (-0.2)(3) \times 0(1)$$

$$net = -0.3$$

$$y = 0 \quad \hat{y} = 0$$

$$e = 0 - 0$$

$$= 0$$

No weight update required.